

Ming-Shan Lee

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Portfolio: <https://atatsuki2000.github.io/Portfolio/>

EDUCATION

University of Southern California (USC) | GPA: 3.8 / 4.0

August 2023 – May 2025

Master of Science

Los Angeles, United States

- Major in Applied Data Science
- Relevant Coursework: Foundations of Data Management, Data Science at Scale, Foundations and Applications of Data Mining

National Tsing Hua University (NTHU) | GPA: 3.8 / 4.0

September 2019 - June 2023

Bachelor of Science

Hsinchu, Taiwan

- Major in Industrial Engineering and Engineering Management
- Relevant Coursework: Discrete Mathematics, Engineering Statistics, Linear Algebra, Data Structures

WORK EXPERIENCE

Micron Technology

July-August 2024

Data Scientist Intern

- Developed and optimized defect prediction machine learning model for semiconductor manufacturing using **Google BigQuery**, achieving **20% improvement in R-squared value** through data mining and factor interaction analysis, with optimized model deployed to production
- **Implemented advanced evaluation metrics and data visualization frameworks** using **Bitbucket** for version control, enabling seamless collaboration and delivering actionable insights to manufacturing stakeholders

NTHU Lean Management Laboratory

May-September 2022

Research Assistant

- Led **machine learning implementation** for semiconductor company's inaugural ML project, strategically securing comprehensive datasets and developing a hybrid system with custom algorithms alongside SVR, Random Forest, and XGBoost models
- **Achieved 25% superior performance** compared to existing implementations across evaluated product lines through pioneering algorithm for imbalanced data challenges in Industry-Academia partnership

Advant Analytics Tactics Ltd.

January-February 2021

Intern

- **Designed optimized scheduling algorithm** using **SQL database analysis** to resolve critical client queue time challenges, creating an innovative approach integrating due date and queue time constraints
- **Eliminated all queue-related issues** while achieving **20% production capacity enhancement**, successfully balancing operational efficiency with customer satisfaction requirements

SKILLS

- Programming Languages: C, C++, Python (NumPy, Pandas, Scikit-learn, PySpark), SQL, HTML, CSS, JavaScript
- Data Management & Visualization: Tableau, MySQL, NoSQL, MongoDB, BigQuery
- Cloud Computing: Google Cloud Platform, AWS

PROJECTS

Better Agent for AppWorld | Python, Large Language Model (LLM), Reinforcement Learning, Fine-Tuning

Aug-Dec 2024

- A project that aims to improve LLM model's performance in AppWorld, a platform that provides LLM agents an environment to test their ability in solving day-to-day tasks.
- GitHub: <https://github.com/Aragath/Better-Agent-For-AppWorld>

NBA Games Real Time Database | Python, MySQL, Flask, HTML, CSS, JavaScript

January-May 2024

- Developed full-stack NBA database application with integrated frontend and backend, providing customized statistics and performance metrics for users
- GitHub: <https://github.com/Atatsuki2000/NBA-Games-Real-Time-Database>